RISE Ethiopia School Survey (Rounds 1 and 2) Data Description

1. Study overview

Research on Improving Systems of Education (RISE) Ethiopia is a longitudinal school survey designed to understand how school resources, teacher capacity, and household context relate to children's foundational learning, with a particular focus on the General Education Quality Improvement Program for Equity (GEQIP-E). The survey collected data at school, teacher, student, and household (caregiver) levels and includes direct assessments of foundational literacy and numeracy for Grade 1 and Grade 4 students.

2. Survey rounds, timing, and panel structure

RISE Ethiopia school survey data covers two rounds:

- **Round 1 (2018/19 academic year):** 168 government primary schools across Addis Ababa, Amhara, Benishangul-Gumuz, Oromia, SNNP, Somali, and Tigray.
- **Round 2 (2020/21 academic year):** 138 schools. Tigray was excluded for security reasons, and some schools in other regions were also inaccessible.

In each round, data were collected in two waves: a baseline at the beginning of the school year (October -November) and an endline at the end of the same school year (May–June). The same sampled children were assessed at baseline and endline within a round, allowing measurement of within-year learning progress. The Round 1 and Round 2 samples are separate cohorts and should not be treated as a single individual-level panel across years.

3. Sampling design and weights

The survey used a multi-stage clustered sampling design. Schools were sampled first, followed by the selection of classes/sections within schools, and then a fixed number of students within selected classes. The selection of schools in each region followed the original RISE Ethiopia research design requirements, which included: (a) incorporating schools from the Young Lives School Surveys (2012-13 and 2016-17); (b) incorporating schools targeted in the first phase of the General Education Quality Improvement Programme for Equity (GEQIP-E) reforms; and

finally, (c) a random selection of other schools to represent both urban and rural populations in each region. In each school, 28 Grade 1 and 28 Grade 4 pupils were randomly selected from up to two classes.

A student-level design weight is provided to adjust for unequal probabilities arising from class/section and student selection within schools (school-size design weight). The weight is the product of: (i) the inverse probability of class selection within the school and (ii) the inverse probability of student selection within the selected class(es). The weight corrects within-school selection only and does not incorporate the probability of school selection at earlier stages or adjustments for non-response/post-stratification.

4. Instruments and levels of data

Each round includes the following instruments (availability varies by wave):

- **Student assessments (Grade 1):** direct one-on-one literacy and numeracy tasks adapted from MELQO and contextualized for Ethiopia. For more on MELQO assessment in Ethiopia, see Kim et al. (2022[1])
- **Student assessments (Grade 4):** language (literacy) and mathematics tests adapted from the Young Lives Ethiopia school survey tools.
- **Socio-Emotional Learning (SEL) survey (Grade 4):** student self-report items using ordered response categories (Likert-type).
- **Household (caregiver) questionnaire:** household roster, socioeconomic conditions, and home learning environment.
- **School principal questionnaire:** school leadership background, staffing, resources, policies, and reform implementation with respect to GEQIP for Equity (GEQIP-E).
- **School facilities observation checklist:** infrastructure and learning resources observed on site.
- **Grade 4 mathematics teacher content knowledge test (endline):** multiple-choice assessment of teacher mathematics knowledge.

[1] Kim, J., Araya, M., Ejigu, C., Hagos, B., Hoddinott, J., Rose, P., Teferra, T., and Woldehanna, T. (2022). Early Learning in Ethiopia: Effects of pre-primary education on school readiness. Early Learning Partnership Ethiopia Phase 2 report. REAL Centre, University of Cambridge, UK

5. Assessment structure and scoring

All learning assessment items have a single correct answer. In addition to the original response or questions, response items are created in the datasets and are coded as binary indicators (1 = correct, 0 = incorrect) for each question across all assessments and rounds. Where a task yields partial points (e.g., some Grade 1 subtasks), raw-score variables are provided alongside item-level variables. Aggregate raw scores and percentage scores are included for quick analysis at the end of data set of the learning assessments.

Grade 1 literacy and numeracy use the same test booklets at baseline and endline within each round. For Grade 4 language and mathematics, 25 items are administered at baseline; however, only 15 of these are retained at endline, while the remaining 10 are replaced with new questions to capture higher-level skills by the end of the year. It is therefore necessary to consult the variable lists in each dataset, as well as the accompanying questionnaires, to confirm which items are common across waves and rounds.

6. Identifiers and how to link files

The datasets are designed to be linked across levels using anonymised numeric identifiers. Key identifiers include:

- **uniqueid:** unique child identifier used to link student-level records across student assessment files, SEL and the household/caregiver file within the same round.
- **school_id:** unique school identifier used across all school, teacher, student, and household-level files.
- **class_id / section** identifiers (where present): identify the sampled class/section within a school.
- **teacher_id (**where present): identifier for teachers (Grade 4 mathematics teacher tested at endline).

Note: Round 1 and Round 2 are separate cohorts. The same numeric value of uniqueid (if it occurs) does not imply the same child across rounds. Analyses comparing 2018/19 and 2020/21 should treat the rounds as repeated cross-sections of cohorts sampled using the same design.

7. Key derived variables

A few derived variables are provided to support analysis and to reduce the need for users to recreate common constructs:

- wi (household wealth index): summary socioeconomic status measure constructed as the simple average of three component indices—housing quality (hq1), consumer durables ownership (cd1), and services quality (sv1). The range is from 0 to 1. Higher values indicate greater relative wealth within the survey sample (do file is attached for this)
- Household size and composition: derived from the household roster, including total household size (hhsize) and sex-by-age group counts (e.g., male05, female05, male612, female612, etc.).
- Assessment totals and percentages: raw-score totals and percent-correct variables for Grade 1 and Grade 4 assessments, plus item-level scoring variables.

Household modules and roster structure: household questionnaire modules have been merged into a single household-level dataset to improve usability.

8. Data files included in the deposit

Files are organized by round (R1 or R2), wave (baseline [B] or endline [E]), and instrument. The file names listed below follow the established naming convention. For each file, the number of cases and the number of variables is documented in the corresponding UK Data Archive data dictionaries provided as attachments.

8.1 Round 1 (2018/19)

Baseline (Oct–Nov 2018):

- RISE R1-B Grade 1 Literacy and Numeracy test.dta
- RISE R1-B Grade 4 student language test.dta
- RISE R1-B Grade 4 student maths test.dta
- RISE R1-B Grades 1 and 4 household questionnaire.dta
- RISE R1-B School principal questionnaire.dta
- RISE R1-B School facilities questionnaire.dta

Endline (May–Jun 2019):

- RISE R1-E Grade 1 Literacy and Numeracy test.dta
- RISE R1-E Grade 4 student language test.dta
- RISE R1-E Grade 4 student maths test.dta
- RISE R1-E Grade 4 Teacher Content Knowledge survey.dta
- RISE R1-E Grade 4 Socio-Emotional Learning survey.dta

8.2 Round 2 (2020/21)

Baseline (Dec 2020–Jan 2021):

- RISE R2-B Grade 1 Literacy and Numeracy test.dta
- RISE R2-B Grade 4 student language test.dta
- RISE R2-B Grade 4 student maths test.dta
- RISE R2-B Grades 1 and 4 household questionnaire.dta
- RISE R2-B School principal questionnaire.dta
- RISE R2-B School facilities questionnaire.dta

Endline (May–Jun 2021):

- RISE R2-E Grade 1 Literacy and Numeracy test.dta
- RISE R2-E Grade 4 student language test.dta
- RISE R2-E Grade 4 student maths test.dta
- RISE R2-E Grade 4 Teacher Content Knowledge survey.dta
- RISE R2-E Grade 4 Socio-Emotional Learning survey.dta